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Thermodynamic assessment of the Co–La and Mo–La systems

C.P. Wang [a], J. Wang [a], X.J. Liu [a,*], I. Ohnuma [b], R. Kainuma [b], K. Ishida [b]

[a] *Department of Materials Science and Engineering, College of Chemistry and Chemical Engineering, Xiamen University, Xiamen 361005, PR China*

[b] *Department of Materials Science, Graduate School of Engineering, Tohoku University, Sendai 980-8579, Japan*

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Abstract

The thermodynamic assessments of the Co–La and Mo–La systems have been carried out by using CALPHAD (Calculation of Phase Diagrams) method on the basis of the experimental data including thermodynamic properties and phase equilibria. Gibbs free energies of the solution phases (liquid, fcc, hcp, bcc and dhcp) were described by subregular solution models with the Redlich–Kister equation, and those of the intermetallic compounds were described by sublattice models. A consistent set of thermodynamic parameters has been derived for describing the Gibbs free energies of each solution phase and intermetallic compound in the Co–La and Mo–La binary systems. The calculated phase diagrams and thermodynamic properties in the Co–La and Mo–La systems are in good agreement with experimental data.

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Keywords: CALPHAD method; Co–La system; Mo–La system; Phase diagram; Thermodynamic assessment

1. Introduction

The Co-based alloys are important materials in applications of magnetic materials and high-temperature alloys [1,2]. The Co–Mo base alloys are potential candidates of ultra-high density recording media because there exists a phase separation between the ferromagnetic fcc phase and paramagnetic fcc phase induced by magnetic transformation [3], same as that in the Co–Cr system [4]. Addition of alloying elements including rare-earth elements is a method to improve the performance of magnetic recording media materials [5,6]. In order to develop the thermodynamic database of phase equilibria in the Co-based alloys for designing new magnetic materials, the thermodynamic assessment of the Co–Mo–La system which forms a part of this thermodynamic database is necessary.

The purpose of this work is to carry out the thermodynamic assessments of the Co–La and Mo–La systems by means of the CALPHAD (calculation of phase diagrams) method, in which the Gibbs free energy of phase is described by a thermodynamic model. The thermodynamic parameters of each phase in the Co–La and Mo–La systems are optimized by fitting the experimental data on thermodynamic properties and phase equilibria.

2. Thermodynamic models

2.1. Solution phases

The Gibbs free energies of the solution phases in an A–B system are described by the following expression:

$$G_{\mathrm{m}}^{\phi} = {}^{0}G_{\mathrm{A}}^{\phi}x_{\mathrm{A}} + {}^{0}G_{\mathrm{B}}^{\phi}x_{\mathrm{B}} + RT(x_{\mathrm{A}}\ln x_{\mathrm{A}} + x_{\mathrm{B}}\ln x_{\mathrm{B}}) + {}^{\mathrm{E}}G^{\phi} + {}^{\mathrm{mg}}G^{\phi} \quad (1)$$

where G_i^{ϕ} is the molar Gibbs free energy of pure element i with the structure ϕ in a nonmagnetic state, which is taken from the compilation by Dinsdale [7], and the term ${}^{\mathrm{E}}G^{\phi}$ is the excess energy, which is expressed in the Redlich–Kister polynomials [8] as

$${}^{\mathrm{E}}G^{\phi} = {}^{0}L_{\mathrm{AB}}^{\phi}x_{\mathrm{A}}x_{\mathrm{B}} + {}^{1}L_{\mathrm{AB}}^{\phi}(x_{\mathrm{A}} - x_{\mathrm{B}})x_{\mathrm{A}}x_{\mathrm{B}} + {}^{2}L_{\mathrm{AB}}^{\phi}(x_{\mathrm{A}} - x_{\mathrm{B}})^{2} \times x_{\mathrm{A}}x_{\mathrm{B}} + \cdots = \sum_{m=0}^{n} {}^{m}L_{\mathrm{AB}}^{\phi}(x_{\mathrm{A}} - x_{\mathrm{B}})^{m}x_{\mathrm{A}}x_{\mathrm{B}} \quad (2)$$

* Corresponding author.
E-mail address: lxj@xmu.edu.cn (X.J. Liu).

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where ${}^{m}L^{\phi}_{AB}$ is the interaction energy between A and B atoms, and expressed as

$$ {}^{m}L^{\phi}_{AB} = a + bT + cT\ln(T) \quad (3) $$

the parameters of a, b and c are evaluated based on the experimental data in the present work.

In the Co–La system, the magnetic contribution to the Gibbs free energies for the fcc, bcc and hcp phases is considered because magnetic transformation occurs in these phases [6]. The term ${}^{mg}G^{\phi}$ is expressed as follows [9]:

$$ {}^{mg}G^{\phi} = RT\ \ln(\beta^{\phi}+1)f(\tau^{\phi}) \quad (4) $$

where β^{ϕ} is a quantity related to the total magnetic entropy, which in most cases is set to the Bohr magnetic moment per mole of atoms; τ^{ϕ} is defined as T/T^{ϕ}_{c}, and T^{ϕ}_{c} is the critical temperature for magnetic ordering, T^{ϕ}_{c} and β^{ϕ} are described by the following expression:

$$ T^{\phi}_{c} = {}^{0}T^{\phi}_{c,Co}x_{Co} + {}^{0}T^{\phi}_{c,La}x_{La} \quad (5) $$

$$ \beta^{\phi} = {}^{0}\beta^{\phi}_{Co}x_{Co} + {}^{0}\beta^{\phi}_{La}x_{La} \quad (6) $$

where ${}^{0}T^{\phi}_{c,Co}$ and ${}^{0}T^{\phi}_{c,La}$ are the curie temperatures for pure Co and La, and ${}^{0}\beta^{\phi}_{Co}$ and ${}^{0}\beta^{\phi}_{La}$ are the Bohr magnetic moments for Co and La, respectively. The $f(\tau^{\phi})$ represents the polynomials obtained by Hillert and Jarl [9] as follows:

$$ ff(\tau^{\phi}) = 1 - \frac{1}{D}\left[\frac{79\tau^{-1}}{140P} + \frac{474}{497}\left(\frac{1}{P}-1\right) \times \left(\frac{\tau^{3}}{6} + \frac{\tau^{9}}{135} + \frac{\tau^{15}}{600}\right)\right] \quad \text{for } \tau \le 1 \quad (7) $$

$$ f(\tau^{\phi}) = -\frac{1}{D}\left(\frac{\tau^{-5}}{10} + \frac{\tau^{-15}}{315} + \frac{\tau^{-25}}{1500}\right) \quad \text{for } \tau > 1 \quad (8) $$

where $D = 518/1125 + 11692/15975(1/P - 1)$, and P depending on the structure is 0.4 for the bcc structure and 0.28 for the others.

2.2. Intermetallic compounds

All intermetallic compounds in the Co–La binary system are treated as stoichiometric phases. The Gibbs free energy for per mole of formula unit A_mB_n can be expressed by the two-sublatice model, as follows:

$$ \Delta G^{A_mB_n}_{f} = {}^{0}G^{A_mB_n}_{f} - m^{0}G^{ref}_{A} - n^{0}G^{ref}_{B} = a' + b'T + c'T\ln T + {}^{mg}G^{A_mB_n} \quad (9) $$

where the $\Delta G^{A_mB_n}_{f}$ denotes the standard Gibbs free energy of formation of the stoichiometric compound from the pure elements. The term ${}^{0}G^{ref}_{i}$ is the molar Gibbs free energy of pure element i (i = A and B, respectively) with its defined reference structure in a nonmagnetic state. The magnetic contribution to the Gibbs free energy ${}^{mg}G^{A_mB_n}$ is similarly given by Eqs. (4)–(8), and the values of β and T_c are taken from the measured critical temperature and the mean Bohr magnetic moment per mole of formula unit. The parameters a', b', and c' are evaluated in the present work.

3. Experimental information

3.1. The Co–La system

The phase diagram in the Co–La system was investigated by several researchers [10–15]. A detailed investigation on the phase diagram in the Co–La system was firstly carried out by Buschow and Velge [13], who measured the phase diagram in the whole composition region by metallography, X-ray diffraction and thermal analyses. The phase stability of the $CoLa_3$, $Co_{23}La_{27}$, Co_3La_2, Co_7La_2, Co_5La and $Co_{13}La$ were determined at 454, 570, 695, 800, 1090, and 1185 °C, respectively. Later, the phase diagram was reinvestigated by Ray [15], who verified that the $Co_{23}La_{27}$, Co_3La_2, Co_7La_2, Co_5La and $Co_{13}La$ form peritectically, and found one compound of the $Co_{19}La_5$, which was not reported in Ref. [13], and determined that the peritectic reaction temperature of the $Co_{19}La_5$ is about 868 °C [15]. By comparing the phase diagrams determined by Buschow and Velge [13] and Ray [15], it is found that the temperatures of the peritectic reactions in the Co-rich portion reported in Ref. [15] are significantly higher than those reported in Ref. [13]. However, Ray [15] did not present the details of his experimental and did not give the exact temperatures of the peritectic reactions and liquidus lines. Ray [15] also stated that the phase diagram in his revision could be viewed as tentative, and many important features remained to be clarified. Therefore, in the present work, the experimental data reported in Ref. [13] are mainly used because Buschow and Velge [13] described experimental procedure and data in detail. The data on the $Co_{19}La_5$ compound reported by Ray [15] are also used. The phase diagram in the Co–La system based on the experimental data in Ref. [13] and Ref. [15] is shown in Fig. 1.

Rezukhina and Kutsev [16] measured the enthalpies and Gibbs free energies of formation of the $Co_{23}La_{27}$, Co_2La_3, Co_7La_2, Co_5La and $Co_{13}La$ compounds at 627 °C by EMF measurements, and Niessen et al. [17] predicted the enthalpies of formation of intermetallic compounds at 627 °C by semi-empirical models based on the energy-band theory. And Nikolaenko et al. [18] measured the mixing enthalpies in liquid of Co–La system at 1550 °C.

The magnetic data of several rare-earth elements and 3d transition elements were compiled by Buschow [6]. The mean values of the critical temperature for magnetic ordering and the Bohr magnetic moment of the $Co_{19}La_5$, Co_7La_2, Co_5La, $Co_{13}La$, and $CoLa_3$ compounds are presented in Table 1. No experimen-

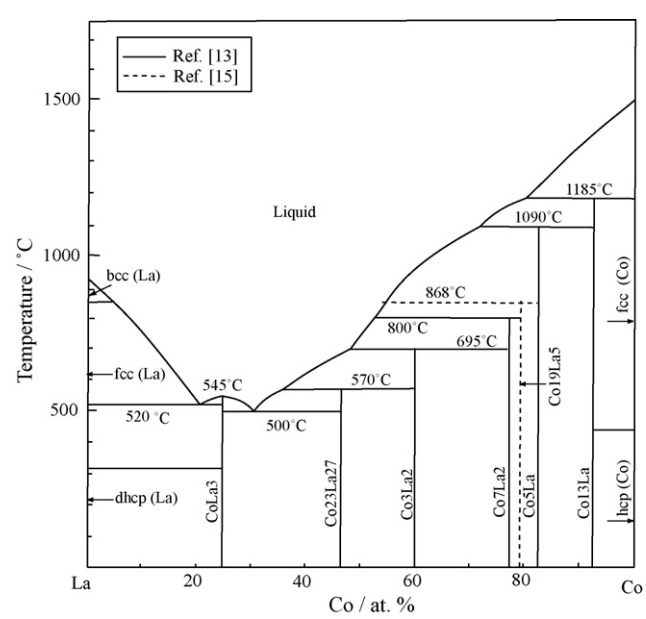

Fig. 1. The experimental Co–La binary phase diagram determined by Buschow and Velge [13] and Ray [15].

tal magnetic data for the $Co_{23}La_{27}$, Co_2La_3 and $CoLa_3$ are available, thus, the magnetic contributions to their Gibbs free energies are ignored in the present work.

3.2. The Mo–La system

The currently accepted Mo–La phase diagram, in which there are five solution phases, namely liquid, bcc, fcc, hcp and dhcp phases, is from Brewer [19] and Okamoto [20], as shown in Fig. 2. Savitskii et al. [21] reported that the solubility of La in the bcc (Mo) phase is less than 0.07 at.%. The important feature in the Mo–La system is that there exists a stable miscibility gap in the liquid phase at high temperatures because of the repulsive interaction between the La and Mo atoms. However, no experimental data on the miscibility gap above 3000 °C are available because of difficulties experimental. Okamoto [20] predicted the critical temperature and composition to be 3500 °C and 65 at.% Mo, respectively. The monotectic reaction of the $L_2 \rightarrow L_1 + \text{bcc (Mo)}$ occurs at 2547 °C, and the corresponding compositions are the 21, 96 and 99.87 at.% Mo, respectively [20].

Table 1
Thermodynamic parameters in the Co–La and Mo–La systems in the present work

Systems	Thermodynamic parameters (J/mol)
Co–La	Liquid phase, format (Co, La) ${}^0L^{liq}_{CoLa} = -35,734 - 0.793 \times T$ ${}^1L^{liq}_{CoLa} = -5150 + 14.181 \times T$ ${}^2L^{liq}_{CoLa} = 6023 - 0.497 \times T$ ${}^3L^{liq}_{CoLa} = 18,450 - 15.756 \times T$ fcc phase, format (Co, La) ${}^0L^{fcc}_{CoLa} = 100,000$ $T^{fcc}_{c,Co} = 1396$ $\beta^{fcc}_{Co} = 1.35$ bcc phase, format (Co, La) ${}^0L^{bcc}_{CoLa} = 60,000$ $T^{bcc}_{c,Co} = 1450$ $\beta^{bcc}_{Co} = 1.35$ $Co_{13}La$ phase, format $(Co)_{0.929}$:$(La)_{0.071}$ $\Delta G^{Co_{13}La}_f = -8589.5 + 3.262 \times T$ $T^{Co_{13}La}_c = 1290$, Ref. [6] $\beta^{Co_{13}La} = 20.50$, Ref. [6] Co_5La phase, format $(Co)_{0.833}$:$(La)_{0.167}$ $\Delta G^{Co_5La}_f = 9434.8 + 1.441 \times T$ $T^{Co_5La}_c = 840$, Ref. [6] $\beta^{Co_5La} = 7.30$, Ref. [6] $Co_{19}La_5$ phase, format $(Co)_{0.792}$:$(La)_{0.208}$ $\Delta G^{Co_{19}La_5}_f = -10729.3 + 1.912 \times T$ $T^{Co_{19}La_5}_c = 616$, Ref. [6] $\beta^{Co_{19}La_5} = 1.93$, Ref. [6] Co_7La_2 phase, format $(Co)_{0.777}$:$(La)_{0.223}$ $\Delta G^{Co_7La_2}_f = -10728.5 + 1.746 \times T$ $T^{Co_7La_2}_c = 490$, Ref. [6] $\beta^{Co_7La_2} = 6.90$, Ref. [6] Co_3La_2 phase, format $(Co)_{0.600}$:$(La)_{0.400}$ $\Delta G^{Co_3La_2}_f = -11550.6 + 1.397 \times T$ $Co_{23}La_{27}$ phase, format $(Co)_{0.460}$:$(La)_{0.540}$ $\Delta G^{Co_{23}La_{27}}_f = -10,857 + 1.495 \times T$ $CoLa_3$ phase, format $(Co)_{0.250}$:$(La)_{0.750}$ $\Delta G^{CoLa_3}_f = -5120 - 2.453 \times T$
Mo–La	Liquid phase, format (Mo, La) ${}^0L^{liq}_{MoLa} = 75,611 - 5.657 \times T$ ${}^1L^{liq}_{MoLa} = -29,422 + 5.301 \times T$ fcc phase, format (Mo, La) ${}^0L^{fcc}_{MoLa} = 50,000$ bcc phase, format (Mo, La) ${}^0L^{bcc}_{MoLa} = 130,000$

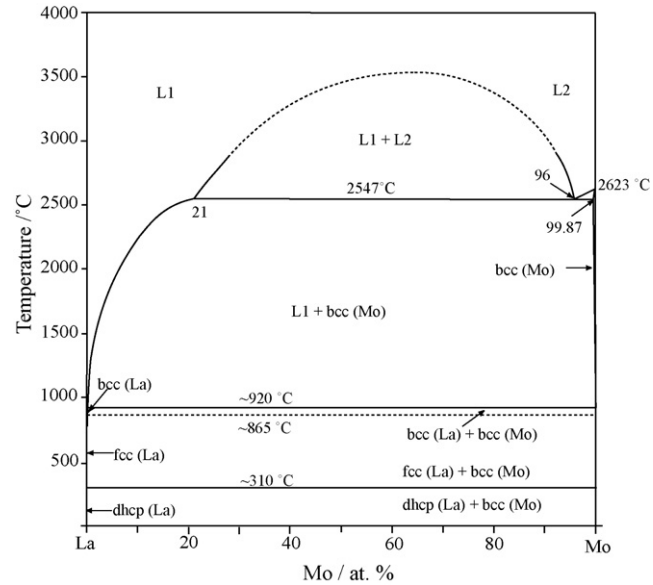

Fig. 2. The phase diagram of the Mo–La binary system reviewed by Okamoto [20].

4. Optimized results and discussion

The optimization was carried out using the PARROT [22] module in the Themo-Calc software [23]. The procedure involves a weighted least-square optimization of the model parameters using experimental information on thermodynamic properties and phase diagram. Each piece of selected information was given a certain weight according to the importance of data, and changed by trial and error during the assessment, until

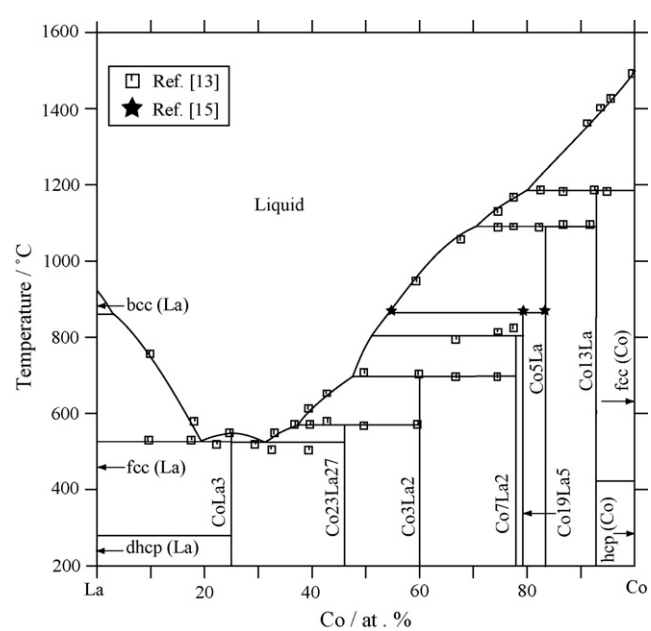

Fig. 3. Calculated phase diagram of the Co–La system with the experimental data [13,15].

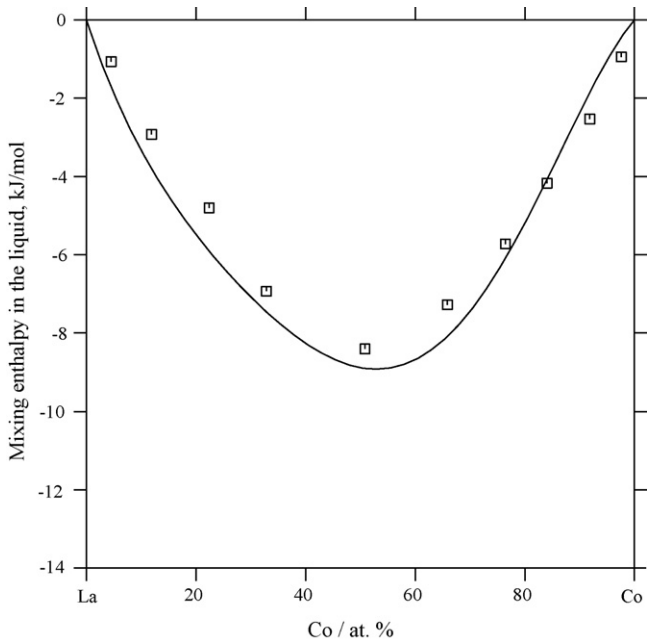

Fig. 4. Calculated mixing enthalpy in liquid at 1550 °C in the Co–La system compared with the experimental data [18].

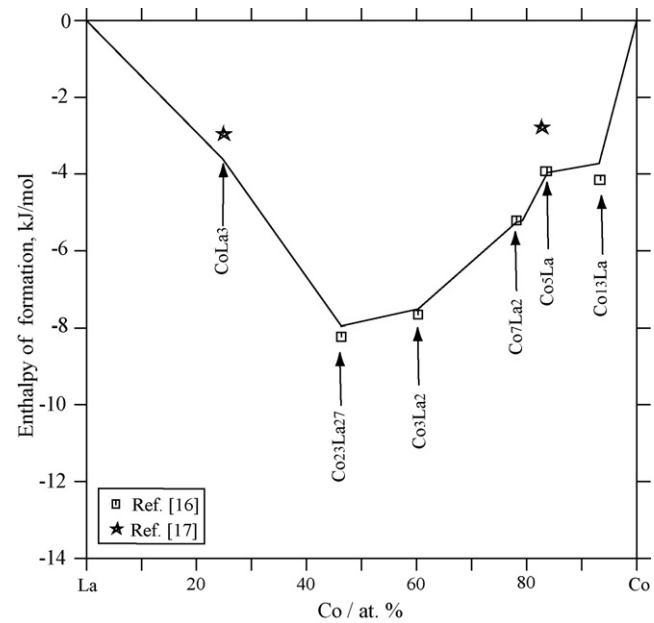

Fig. 6. Calculated enthalpy of formation of intermetallic compounds at 627 °C in the Co–La system compared with the experimental data [16,17].

most of the selected experimental information is reproduced within the expected uncertainty limits.

4.1. The Co–La system

The calculated phase diagram in the Co–La system is shown in Fig. 3, comparing with all experimental data used in the present optimization. It is seen from Fig. 3 that the calculated results are in good agreement with the experimental data reported by Buschow and Velge [13] and Ray [15]. The calculated mix enthalpy in the liquid phase and enthalpies and Gibbs free energies of formation for the compounds are plotted in Figs. 4–6, respectively. A agreement between the calculated and experimental results [15,16] is obtained. The thermodynamic parameters optimized in the present work are listed in Table 1.

All invariant equilibria in the Co–La system are summarized in Table 2, in which the experimental data are also listed for comparison [13,15]. The values of the calculated congruent melting temperature of $LaCo_3$ and the peritectic temperatures are in agreement with the corresponding experimental data [13,15].

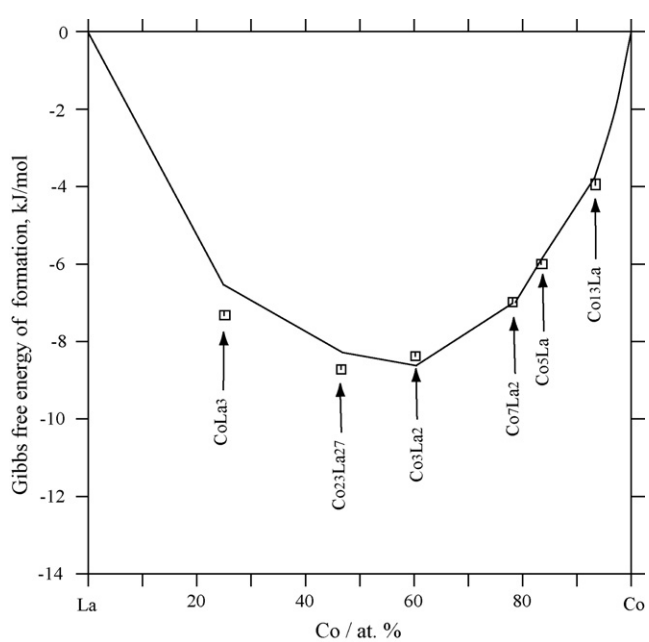

Fig. 5. Calculated free Gibbs energy of formation of intermetallic compounds at 627 °C in the Co–La system compared with the experimental data [16].

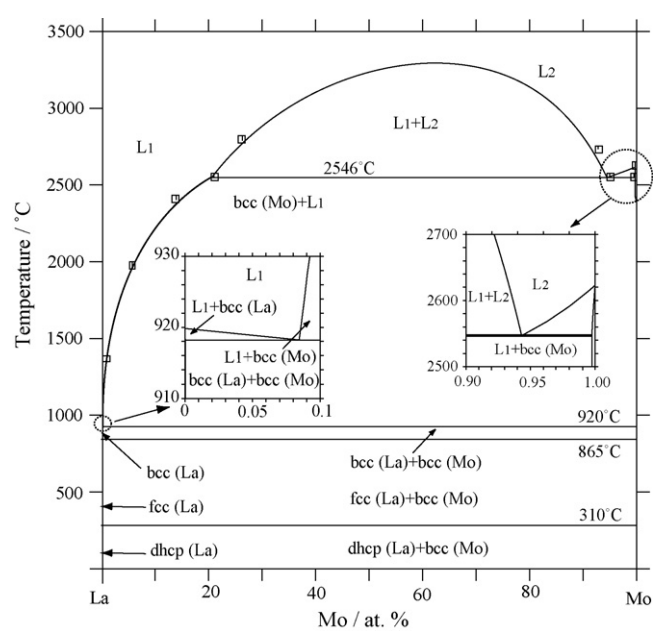

Fig. 7. Calculated phase diagram of the Mo–La system compared with experimental data [20].

Table 2
Invariant reactions in the Co–La system

Reaction	Composition (at.% Co)			Temperature (°C)	Reference
L + fcc (Co) → $Co_{13}La$	80.3	0.1	92.9	1185	[13]
	81	0.1	92.9	1275	[15]
	78.9	0.1	92.9	1185	Present work
L + $Co_{13}La$ → Co_5La	71.7	92.9	83.4	1090	[13]
	73	92.9	83.4	1124	[15]
	69.9	92.9	83.4	1090	Present work
L + Co_5La → $Co_{19}La_5$	–	–	–	–	[13]
	55.2	83.4	79.1	868	[15]
	55.7	83.4	79.1	868	Present work
L + $Co_{19}La_5$ → Co_7La_2	52.8	79.1	77.8	800	[13]
	54.3	79.1	77.8	843	[15]
	52.3	79.1	77.8	800	Present work
L + Co_7La_2 → Co_3La_2	48.1	77.8	60	695	[13]
	49.5	77.8	60	727	[15]
	47.6	77.8	60	695	Present work
L + Co_3La_2 → $Co_{23}La_{27}$	35.7	60	46	570	[13]
	37.5	60	46	596	[15]
	37.6	60	46	570	Present work
L → $Co_{23}La_{27}$ + $CoLa_3$	30.6	46	25	500	[13]
	~31	46	25	500	[15]
	31.5	46	25	523	Present work
L → fcc (La) + $CoLa_3$	20.5	0.1	25	520	[13]
	~20	0.1	25	520	[15]
	18.7	0.1	25	521	Present work
L → $CoLa_3$	–	25	–	545	[13]
	–	25	–	545	[15]
	–	25	–	543	Present work

Table 3
Invariant reactions in the Mo–La system

Reaction	Composition (at.% Mo)			Temperature (°C)	Reference
L_2 → L_1 + bcc (Mo)	96	21	99.87	2547	[19]
	95	21	99.87	2546	Present work
L_1 → bcc (La) + bcc (Mo)	0.090	0	100	918	[19]
	0.085		100	917	Present work

4.2. The Mo–La system

Since no experimental thermodynamic data in the Mo–La binary system are available, in the present work, the optimization was carried out based on the experimental data of phase diagram reviewed by Okamoto [20]. The calculated phase diagram with experimental data is shown in Fig. 7. All invariant equilibria in the Mo–La system are summarized in Table 3. The calculated results are in an agreement with the experimental data by Brewer [19] and Okamoto [20]. The calculated results indicate that the monotectic reaction temperature of the L_2 → L_1 + bcc (Mo) is 2546 °C, and the corresponding compositions are 21.96 and 99.87 at.% Mo, respectively. The calculated critical temperature and composition of the liquid miscibility gap are 3380 °C and 64 at.% Mo in the present work. All thermodynamic parameters optimized in the present work are listed in Table 1.

5. Conclusions

The phase diagrams and thermodynamic properties in the Co–La and Mo–La systems were evaluated by combining the thermodynamic models and the available experimental information. A consistent set of thermodynamic parameters has been derived for describing the Gibbs free energies of each solution phase and intermetallic compounds in the Co–La and Mo–La binary systems, leading to a good agreement between the calculated results and most of the experiment data found in literature.

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